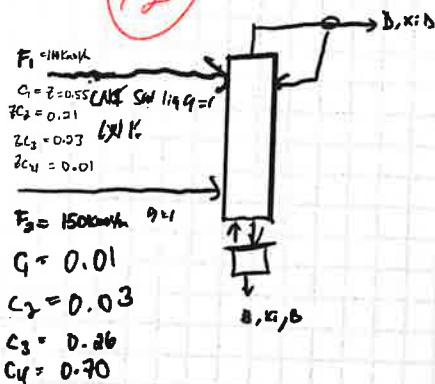


HW 4

DIANA RODRIGUEZ 313

D5)

7.5



$$\sum L_{C3} = 99.3 \text{ UK}$$

$$\sum L_{C4} = 99.5 \text{ HK}$$

$$C_1: \sum x_{C1,D} = (F_1 \cdot z_{C1,1}) + (F_2 \cdot z_{C1,2}) = 100(0.55) + 150(0.01) = 56.5 \frac{\text{kmol}}{\text{h}}$$

$$C_2: \sum x_{C2,D} = (F_1 \cdot z_{C2,1}) + (F_2 \cdot z_{C2,2}) = 100(0.21) + 150(0.03) = 25.5 \frac{\text{kmol}}{\text{h}}$$

$$C_3: \sum x_{C3,D} = \frac{D \cdot x_{C3,D}}{F_1 \cdot z_{C3,1} + F_2 \cdot z_{C3,2}} = 0.993 \Rightarrow \sum x_{C3,D} = (0.993)(100 \cdot 0.23 + 150 \cdot 0.26) = 61.556$$

$$C_4: \sum x_{C4,D} = (1 - \sum x_{C1,D}) \Rightarrow 0.005 = \frac{D \cdot x_{C4,D}}{F_1 \cdot z_{C4,1} + F_2 \cdot z_{C4,2}} \Rightarrow \sum x_{C4,D} = 0.005(100 \cdot 0.01 + 150 \cdot 0.7) = 0.53 \frac{\text{kmol}}{\text{h}}$$

Mole fraction of finding D

$$x_{i,D} = \frac{x_{i,D}}{D}$$

$$D = 56.5 + 25.5 + 61.556 + 0.53 = 144.096 \frac{\text{kmol}}{\text{h}}$$

$$x_{C1,D} = \frac{D \cdot x_{C1,D}}{D} = \frac{56.5}{144.096} = 0.39$$

$$x_{C2,D} = \frac{D \cdot x_{C2,D}}{D} = \frac{25.5}{144.096} = 0.18$$

$$x_{C3,D} = \frac{D \cdot x_{C3,D}}{D} = \frac{61.556}{144.096} = 0.43$$

$$x_{C4,D} = \frac{D \cdot x_{C4,D}}{D} = \frac{0.53}{144.096} = 0.004$$

Again for bottom

$$C_1: \sum x_{C1,B} = 1 \Rightarrow \sum x_{C1,B} = 0$$

$$C_2: \sum x_{C2,B} = 1 \Rightarrow \sum x_{C2,B} = 0$$

$$C_3: \sum x_{C3,B} = \frac{B \cdot x_{C3,B}}{F_1 \cdot z_{C3,1}}$$

$$(1 - 0.993) = \frac{B \cdot x_{C3,B}}{100(0.23) + 150(0.26)} \Rightarrow \sum x_{C3,B} = 0.007(100 \cdot 0.23 + 150 \cdot 0.26) = 0.434 \frac{\text{kmol}}{\text{h}}$$

$$C_4: \sum x_{C4,B} = \sum x_{C4,B} = (F_1 \cdot z_{C4,1}) + (F_2 \cdot z_{C4,2}) = 0.995(100 \cdot 0.01 + 150 \cdot 0.7)$$

$$B = 105.904$$

$$x_{C1,B} = \frac{B \cdot x_{C1,B}}{B} = 0$$

$$x_{C2,B} = \frac{B \cdot x_{C2,B}}{B} = 0$$

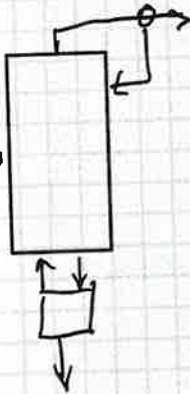
$$x_{C3,B} = \frac{B \cdot x_{C3,B}}{B} = 0$$

$$x_{C3,B} = \frac{0.434}{105.904} = 0.004$$

$$x_{C4,B} = \frac{105.904}{105.904} = 0.995$$

D2

$T_1 = 84^\circ\text{C}$
 $K_A = 1$
 $K_B = 0.84$
 $K_C = 0.46$
 $T_2 = 40^\circ\text{C}$
 $K_A = 1.13$
 $K_B = 1.00$
 $K_C = 0.46$



$$f_{LK} = 0.99 \text{ LK A}$$

$$f_{HK} = 0.950 \text{ HK B}$$

$$f_{LK} = 0.999 \text{ HK C}$$

$$84^\circ\text{C} \propto = \frac{1.00}{0.46} = 2.38$$

$$40^\circ\text{C} \propto = \frac{1.13}{0.46} = 2.44$$

Underwood eqns

$$\bar{\alpha}_{LK} = \frac{1}{\alpha_{LK}} = \frac{1}{2.38} = 0.42$$

$$N_{min} = \ln \left[\frac{f_{LK,D} (1 - f_{LK,B})}{(1 - f_{LK,D}) f_{LK,B}} \right]$$

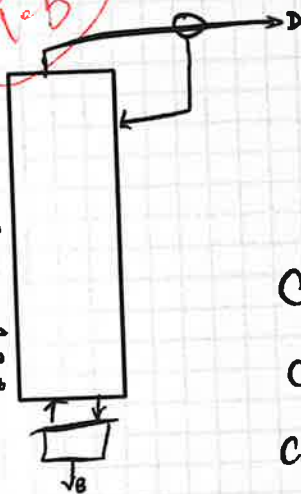
$$= \ln \left[\frac{0.999 (1 - 0.999)}{(1 - 0.999) 0.999} \right]$$

$$N_{min} = 15.6$$

16 stages

D7

$F = 155.0 \text{ kmol/hr}$
 $C_1 = C_2 = 0.05$
 $C_3 = C_4 = 0.10$
 $C_5 = C_6 = 0.20$
 $C_7 = C_8 = 0.20$
 $C_9 = C_{10} = 0.20$



$$f_{LK} = 0.99 \text{ LK C4}$$

$$f_{HK} = 0.983 \text{ HK C8}$$

Distillate

$$C_1: DX_{C1} = f_{LK} \cdot (C_1 + F) = 7.75 \text{ kmol/hr}$$

$$DX_{C1} = (0.05)(155.0) = 7.75 \text{ kmol/hr}$$

$$C_2: DX_{C2} = (0.10)(155.0) = 15.5 \text{ kmol/hr}$$

$$C_4: DX_{C4} = (0.99)(0.15)(155.0) = 23.018 \text{ kmol/hr}$$

$$C_5: DX_{C5} = (1 - 0.983)(0.20)(155.0) = 0.579 \text{ kmol/hr}$$

$$C_6: DX_{C6} = 0$$

$$C_9: DX_{C9} = 0$$

Key components

- C1 LNK
- C2 LNK
- C4 LK
- C5 HK
- C6 HNK
- C9 HNK

$$D = 46.8472 \text{ kmol/hr}$$

Bottom

$$Bx_{C1} = 0$$

$$Bx_{C2} = 0$$

$$Bx_{C4} = \int_{2C4} \cdot (F_{C4} \cdot z_{C4})$$

$$= (1 - 0.99)(1550 \cdot 0.15) = 0.2325 \frac{\text{kmol}}{\text{h}}$$

$$Bx_{C5} = 0.983(155 \cdot 0.26) = 40.3 \frac{\text{kmol}}{\text{h}}$$

$$Bx_{C6} = (155 \cdot 0.22) = 34.1 \frac{\text{kmol}}{\text{h}}$$

$$Bx_{C7} = (155 \cdot 0.26) = 40.3 \frac{\text{kmol}}{\text{h}}$$

$$B = 109.15 \frac{\text{kmol}}{\text{h}}$$

Mole fractions for Distillate

$$x_{C1,D} = \frac{7.75}{46.7472} = 0.16$$

$$x_{C2,D} = \frac{15.5}{46.7472} = 0.33$$

$$x_{C4,D} = \frac{23.0195}{46.7472} = 0.49$$

$$x_{C5,D} = \frac{0.5797}{46.7472} = 0.01$$

$$x_{C6,D} = 0$$

$$x_{C7,D} = 0$$

Mole fractions for Bottoms

$$x_{C1,B} = 0$$

$$x_{C2,B} = 0$$

$$x_{C4,B} = \frac{0.2325}{108.1528} = 0.002$$

$$x_{C5,B} = \frac{38.5207}{108.1528} = 0.31$$

$$x_{C6,B} = \frac{34.1}{108.1528} = 0.33$$

$$x_{C7,B} = \frac{40.3}{108.1528} = 0.37$$

b) $k_a = 95$ $\alpha_a = 452$

$k_b = 7.9$ $\alpha_b = 35.9$

$k_c = 2.3$ $\alpha_c = 10.5$

$k_d = 0.22$ $\alpha_d = 0.626$

$k_f = 0.11$ $\alpha_f = 0.400$

$k_f = 0.088$

Distillate Bubble pt Temp, $1 = \sum \frac{x_{i,D}}{k_i(T)} = 36^\circ\text{C}$

yes, it will be expensive due to necessary Δ in temp to cool

c) $N_{min} = \frac{\ln \left[\left(\frac{0.99}{0.01} \right) \cdot \left(\frac{0.183}{0.017} \right) \right]}{\ln(10.5)} = 3.68$

$N_{min} = 4$ total stages

$\left(\frac{L}{V} \right)_{min} = \left(\frac{V-D}{D} \right)_{min}$

$V_{min} = \sum \frac{\alpha_{i-r} \cdot (x_{i,D} - D)}{\alpha_{i-r} - \phi}$

$D = \frac{(432)(7.75)}{432 - \phi} + \frac{(35.9)(15.5)}{35.9 - \phi} + \frac{(10.5)(23.25)}{10.5 - \phi} + \frac{(1)(34.1)}{1 - \phi} + \frac{(0.626)(24.1)}{0.626 - \phi} + \frac{(0.400)(40.3)}{0.400 - \phi}$

$\phi = 2.143$ ($\alpha_{C6} < \phi < \alpha_{C7}$)

$V_{min} = \sum \frac{\alpha_{i-r} \cdot (x_{i,D} - D)}{\alpha_{i-r} - \phi}$

$V_{min} = \frac{(432)(7.75)}{(432 - 2.143)} + \frac{(35.9)(15.5)}{35.9 - 2.143} + \frac{(10.5)(23.25)}{10.5 - 2.143} + \frac{(1)(34.1)}{1 - 2.143} = 59.9 \frac{\text{kmol}}{\text{h}}$

$$d) \left(\frac{L}{D}\right)_{\min} = \frac{V_{\min} - D}{D} = \frac{50.09 - 46.85}{46.85}$$

$$\left(\frac{L}{D}\right)_{\min} = 0.125$$

e) When $M = 1.10$

$$\frac{L}{D} = (1.10)(0.125) = 0.13$$

$$X = \frac{V/D - (L/D)_{\min}}{(L/D) + 1} = \frac{0.143 - 0.125}{0.173 + 1} = 0.0114$$

$$Y = \frac{1 - X^{0.0081}}{1 - 0.99357 X^{0.0081}} = 0.785$$

$$N = \frac{(Y + N_{\min})}{(1 - Y)} \quad N = 15.7 \approx 16 \text{ stages}$$

When $M = 1.10$

$$\frac{L}{D} = 0.1375$$

$$X = 0.011$$

$$Y = 0.686$$

$N = 15 \text{ stages}$

When $M = 2.0$

$$\frac{L}{D} = 0.25$$

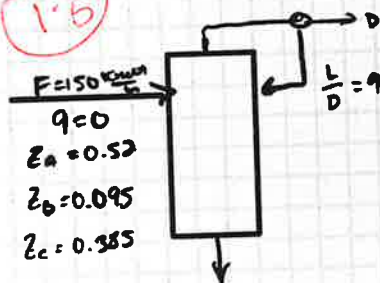
$$X = 0.1$$

$$Y = 0.527$$

$N = 10 \text{ stages}$

f) An alternative to distillation may be chosen because I found the distillation to require many stages, which will be costly

D9 1.6



A Benzene: 1K
B Toluene: HK
C Xylene: HK

$$\alpha_{AB} = 2.22 \Rightarrow \frac{1}{2.09}$$

$$\alpha_{BC} = 2.01$$

$$\alpha_{AC} = 0.178$$

a)

Distillate:

$$x_{A,D} = 0.993$$

$$x_{B,D} = 0.007$$

$$x_{C,D} = 0$$

$$F z_A = D x_{A,D} + B x_{A,B}$$

$$(150)(0.52) = D(0.993) + 78.55$$

Mass balance

$$F = D + B$$

$$F z_B = D x_{B,D} + B x_{B,B}$$

$$(150)(0.095) = (78.55)(0.007) + 71.45 x_{B,B}$$

$$x_{B,B} = 0.19$$

$$x_{A,B} = 1 - 0.19 = 0.81$$

No HK in distillation

$$x_{C,D} = 0$$

$$x_{C,B} = 1 - 0.007 = 0.993$$

$$x_{C,D}$$

$$G = F z_C = D x_{C,D} + B x_{C,B}$$

$$150(0.385) = B(0.993) \Rightarrow B = 98.55 \frac{\text{kmol}}{\text{h}}$$

$$b) N_{min} = \frac{\ln \left(\frac{(\frac{x_{c,D}}{x_{c,D}})_0}{(\frac{x_{c,D}}{x_{c,D}})_0} \right)}{\ln \alpha_{LK-HK} (\alpha_{ST})} = \frac{\ln \left[\frac{(0.993/0.007)}{(0.192/0.192)} \right]}{\ln (2.22)} = 18.58$$

$N_{min} = 19 \text{ stages}$

$$c) N_{min} = \frac{\ln \left[\left(\frac{x_{c,D}}{x_{c,D}} \right)_0 / \left(\frac{x_{c,D}}{x_{c,D}} \right)_0 \right]}{\ln \alpha_{LK-HK}} = \frac{\ln \left[\frac{(0.007/0.192)}{(0.192/0.192)} \right]}{\ln (2.01)} = 19$$

$$x_{c,D} = \frac{0.007}{(0.192/0.192)} \rightarrow x_{c,D} = 5.4 \times 10^{-4}$$

$$d) F(1-q) = \sum \frac{d_{i-1} \cdot F \cdot z_i}{d_{i-1} - \phi}$$

$$150 = \frac{(2.22)(150)(0.5)}{2.22 - \phi} + \frac{(150)(0.095)}{1 - \phi} + \frac{(0.498)(150)(0.385)}{2.498 - \phi}$$

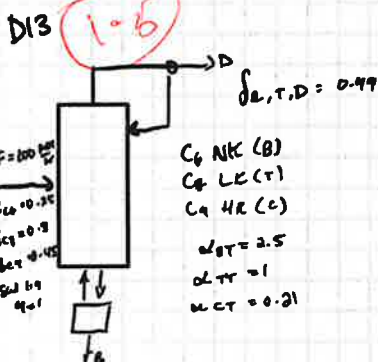
$$150(2.22 - \phi)(1 - \phi)(0.498 - \phi) = 173.6(1 - \phi)(0.498 - \phi) + 14.25(2.22 - \phi)(0.498 - \phi) + 28.7595(2.22 - \phi)(1 - \phi)$$

$$\phi = 1.42$$

$$Y_{min} = \frac{2.22 - 1.42}{2.22 - 1.42} + \frac{(0.498)(4 \times 10^{-4})}{0.498 - 1.42} = 214.32 \frac{\text{mol}}{\text{h}} \Rightarrow \frac{L}{D_{min}} = \frac{214.32 - 77.57}{77.57} = 1.73$$

$$e) x = \frac{4 - 1.13}{4 + 1} = 0.73$$

$$y = \frac{1 - 0.73}{1 - 0.9425(0.73)^{0.0031}} = 0.13 \quad N = \frac{0.12 + 1.9}{1 - 0.13} = 21.79 \approx 22 \text{ stages}$$



$$a) \int_{0.95}^{0.99} \frac{D \pi_i D}{F \cdot z_i}$$

Distillation

$$D \cdot x_{c,D} = (100 \cdot 0.35) \frac{25 \text{ kmol}}{\text{h}}$$

$$D \cdot x_{T,D} = 0.99(100 \cdot 0.3) = 29.7 \frac{\text{kmol}}{\text{h}}$$

$$D \cdot x_{B,D} = (1 - 0.99)(100 \cdot 0.45) = 0.9 \frac{\text{kmol}}{\text{h}}$$

$$D = 25 + 29.7 + 0.9 = 55.6 \frac{\text{kmol}}{\text{h}}$$

$$D = \frac{25 + 29.7}{1.5 - \phi} + \frac{1 - \phi}{1 - \phi} + \frac{0.91 \cdot 25}{0.21 - \phi}$$

$$0 = 24.25(1 - \phi)(0.21 - \phi) + 0.9(2.5 - \phi)(0.21 - \phi) + 5.25(2.5 - \phi)(1 - \phi)$$

$$0 = 15.375 - 74.25\phi - 15.375\phi + 74.25\phi^2 + 0.4725 - 2.25\phi - 0.189\phi + 0.9\phi^2 + 13.725 - 12.125\phi - 5.25\phi + 5.25\phi^2$$

$$D = 80.4\phi^2 - 110.65\phi + 29.19 \quad \phi = 1.58$$

$$Y_{min} = \frac{(2.5)(29.7)}{(2.5 - 1.58)} + \frac{(1)(0.9)}{1 - 1.58} + \frac{(0.21)(25)}{0.21 - 1.58} = 75.32$$

$$\left(\frac{L}{D} \right)_{min} = \frac{75.32 - 55.6}{55.6} = 0.355 \quad N_{min} = \frac{\ln \left(\frac{(0.99/1-0.99)}{(1-0.98/0.98)} \right)}{\ln (2.21)} \approx 6 \text{ stages}$$

$$Y = \frac{N - N_{\min}}{N+1} = \frac{9-6}{10} = 0.3$$

$$X = \left(\frac{1-0.3}{1-0.992(0.3)} \right)^{1/0.0031} = 0.41$$

$$0.41 = \frac{\frac{b}{b+1} - 0.355}{\frac{b}{b+1}} \Rightarrow 0.41 \frac{b}{b+1} + 0.41 \Rightarrow \frac{b}{b+1} = 0.355 \Rightarrow \frac{b}{b+1} = 0.297$$

$$b) D = \frac{2.25 - 19.7}{2.25 - \phi} + \frac{0.9}{1 - \phi} + \frac{5.25}{0.21 - \phi} \Rightarrow D = 14.633K - 66.825\phi - 14.05535\phi + 66.825\phi^2 + 0.92535 - 1.025\phi - 0.179\phi + 0.9\phi^2 + 11.1725 - 11.3125\phi - 5.35\phi + 5.35\phi^2$$

$$\phi = 1.51$$

$$V_{\text{an}} = \frac{66.825}{(2.25)(1.51)} + \frac{0.9}{1-1.51} + \frac{5.25}{0.21-1.51} = 94.5$$

$$\left(\frac{L}{D} \right)_{\min} = \frac{94.5 - 55.6}{55.6} = 0.52$$

$$0.41 = \frac{\frac{b}{b+1} - 0.520}{\frac{b}{b+1}} \Rightarrow 0.41 \frac{b}{b+1} + 0.41 \Rightarrow \frac{b}{b+1} = 0.520$$

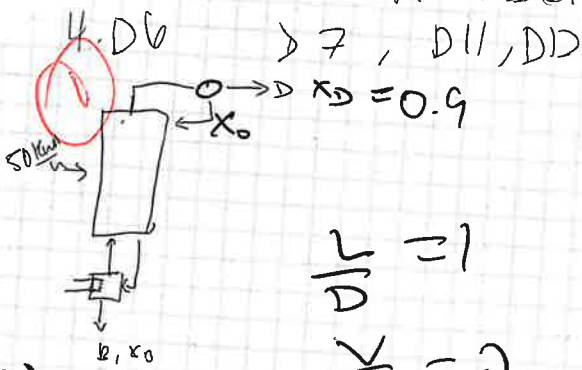
$$\frac{b}{b+1} = 0.520$$

$$= 1.6$$

DIATA RODRIGUEZ

Assignment 4

4



$$\frac{L}{V} = \frac{1}{2}$$

$$\frac{L}{D} = 1$$

$$\frac{V}{B} = 2$$

$$\left(\frac{1}{2}\right) 0.9$$

a)

Enriching

$$y = \frac{L}{V}x + \left(1 - \frac{L}{V}\right)x_0$$

$$V = L + D$$

$$\frac{L}{V} = \frac{L}{L+D} = \frac{0.9}{0.9+1} = \frac{1}{2} = 0.5$$

$$x_0 \left(1 - \frac{L}{V}\right) = 0.5$$

$$(1 - 0.5) 0.9 =$$

$$y_1 = x_0 = x_0 = 0.9$$

$$(0.9, 0.9)$$

$$(0, 0.4)$$

$$\frac{L}{V} = \frac{0.9}{0.9+1} = \frac{1}{1.1} = 0.5$$

$$y = - (0.5 - 1) 0.9 =$$

$$- (-0.45)$$

$$y = 0.5x + 0.45$$

$$(0.9, 0.9)$$

stripping

$$y = \frac{L}{V}x - \left(\frac{L}{V} - 1\right)x_0$$

$$\frac{L}{V} = \frac{V+B}{V} = \frac{2+1}{2}$$

$$y = \frac{3}{2}x - 0.025$$

Doesn't cross

$$\left(\frac{3}{2} - 1\right) 0.05$$

$$y = \frac{V}{B}x - \left(\frac{V}{B} - 1\right)x_0$$

$$y = 2x - (2 - 1) 0.05$$

$$y = 2x - 0.05$$

$$(0.05, 0.05)$$

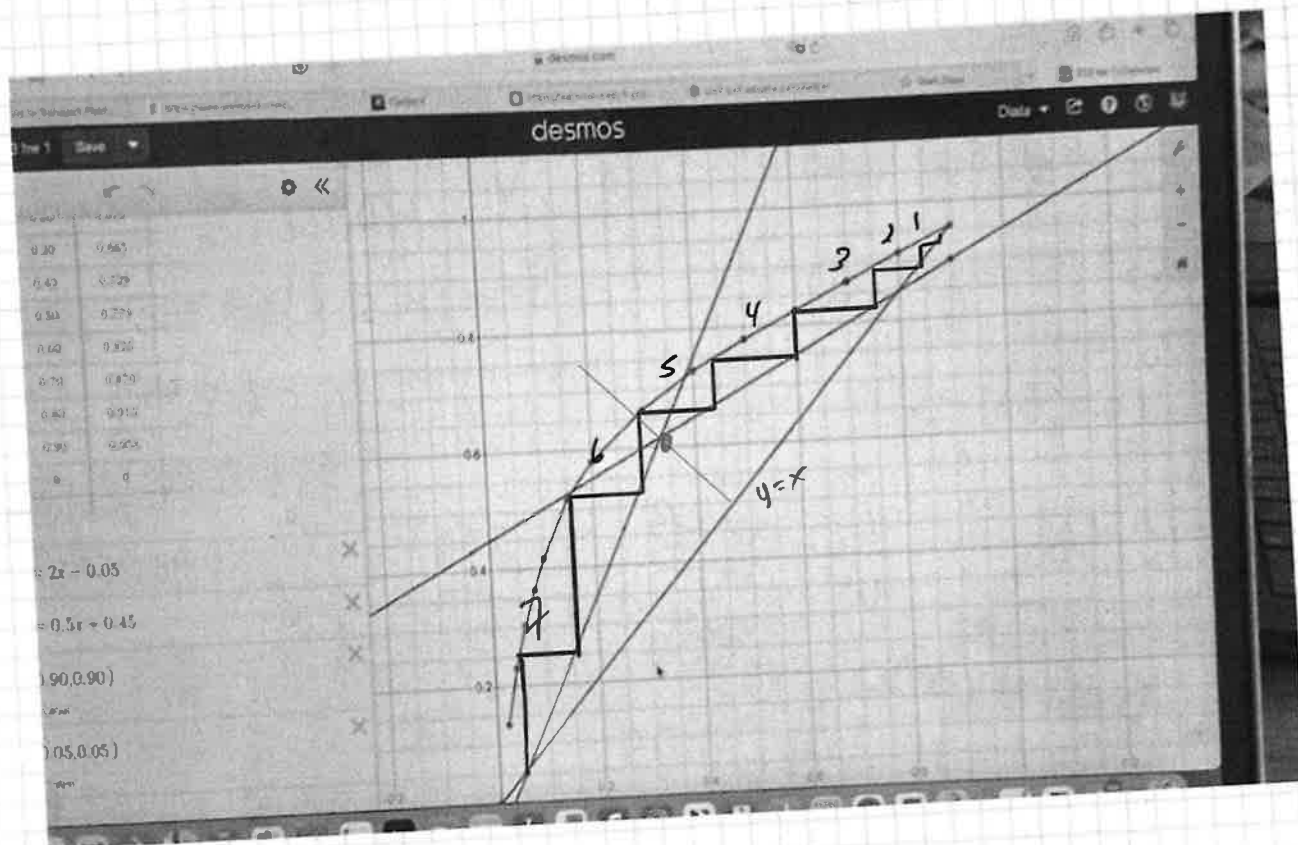
c) $q=1$

feed location is where the two operating lines intersect.

Intersect
at

$x = 0.833$

$z_f = 0.833$



b) optimum stage = 5
Total # of stages needed
= 7

D7.* A distillation column with a total condenser is separating a feed that is 30 mol% acetone from ethanol. Distillate $x_D = 0.90$, and bottoms $x_B = 0.13$, both mole fractions acetone. Assume CMO is valid. $L/V = 0.8$. Boilup ratio is $v/B = 1.0$.

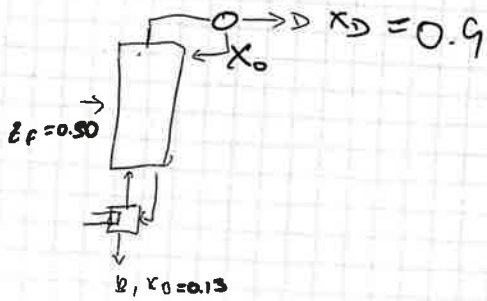
- Find the composition of the liquid leaving the fifth stage below the total condenser.
- The column has a partial reboiler that acts as an equilibrium contact. Find the vapor composition leaving the second stage above the partial reboiler.
- Find optimum feed plate location, total number of stages, and required q value of feed.

Equilibrium data for acetone (A) and ethanol at 1 atm (Perry et al., 1963, pp. 13-4):

x_A	0	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.50	0.60	0.70	0.80	0.90	1.0
y_A	0	0.262	0.348	0.417	0.478	0.524	0.566	0.605	0.674	0.739	0.802	0.865	0.929	1.0
$T^{\circ}\text{C}$	78.3	-	-	-	67.3	65.9	-	63.6	-	60.4	59.1	58.0	57.0	56.1

3) 3

193



$$\frac{L}{V} = \frac{z_F}{x_D} = 0.56$$

$$\frac{L}{V} = 1.0$$

Enriching

$$y = \frac{L}{V} x + (1 - \frac{L}{V}) x_D$$

$$y = 0.56x + (1 - 0.56)0.9$$

$$y = 0.56x + 0.48$$

$$y_1 = x_D = x_0 = 0.9 \quad (0.9, 0.9)$$

stripping

$$y = \frac{L}{V} x + (\frac{L}{V} - 1) x_B \quad \frac{L}{V} = 1$$

$$\frac{L}{V} = \frac{2+1}{1/0.9} = 2$$

$$y_{m1} = (\frac{L}{V} - 1) x_B = (2 - 1)(0.13) = 0.13$$

$$z_F = 0.50$$

$$y = 2x - 0.13$$

where they intersect

start $\mathcal{T}$

a) $x_5 = 0.53$

b) $y_2 = 0.52$

c) optimum feed location: $\mathcal{T}$

9 stages $(0.3, 0.3)$

$q = 0.67$

$z = x = y$

Intersection at

$x = 0.258 \quad y = 0.38$

Slope-

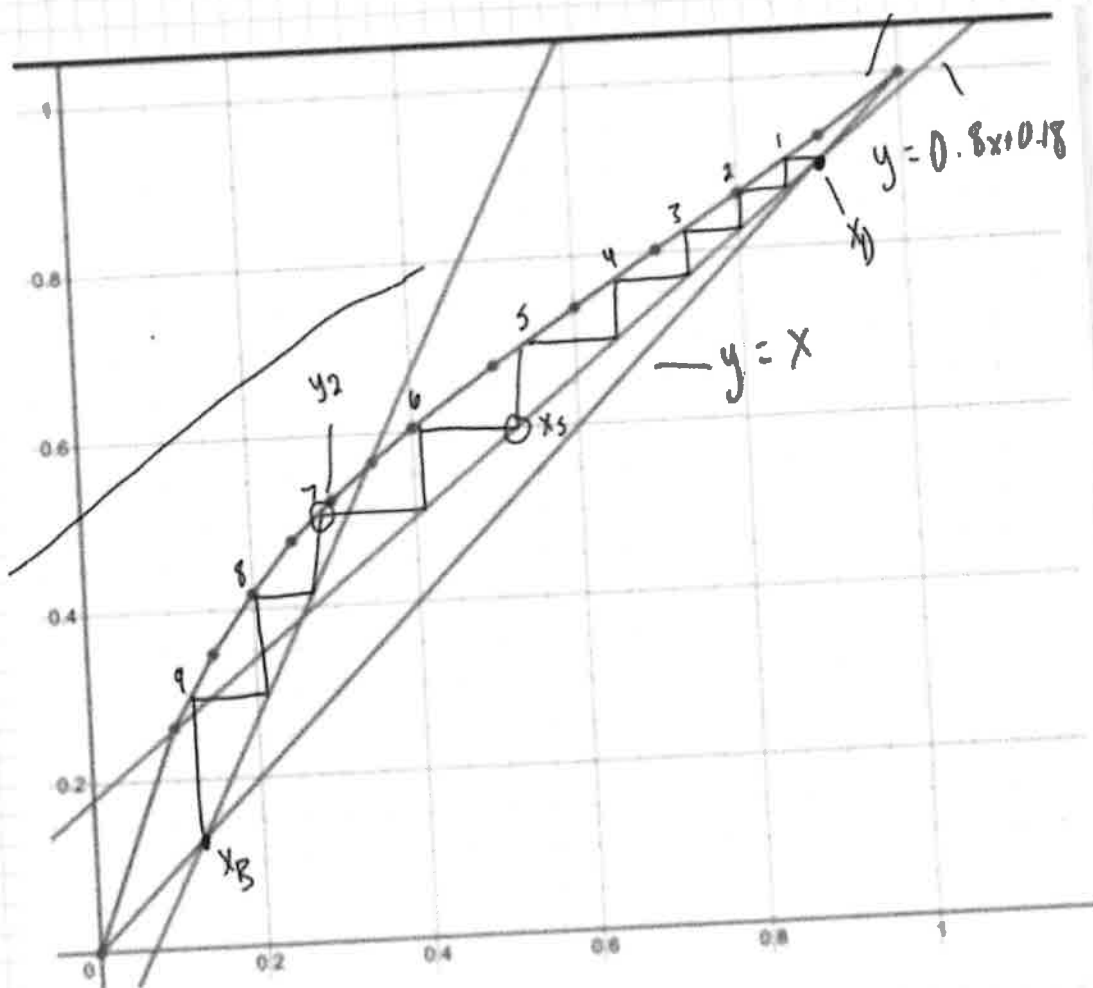
$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{0.38 - 0.3}{0.25 - 0.3} = -2.06$$

$$\frac{q}{q-1} = -2.06$$

$$q = -2.06q + 2.06$$

$$q = 0.67$$

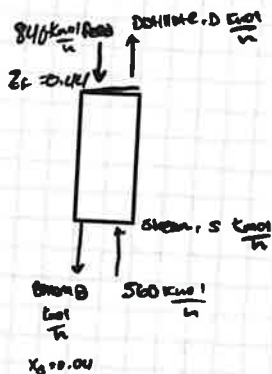
pg 4



D11. A beer still (see Figure 3-9 in Problem 3.E2) is being used to separate ethanol from water at 1.0 atm. The saturated liquid feed flow rate is $F = 840.0 \text{ kmol/h}$. The feed is 44.0 mol% ethanol. The saturated vapor steam is pure water with ratio of steam flow rate S to feed rate, $S/F = 2/3$. We desire a bottoms product that is 4.0 mol% ethanol. CMO is valid. Equilibrium data are in Table 2-1.

- Find the mole fraction of ethanol in the distillate vapor, y_D .
- Find the number of equilibrium stages required.
- If the feed is unchanged and the S/F ratio is unchanged, but the number of stages is increased to a very large number, what is the lowest bottoms mole fraction of ethanol that can be obtained?

P35



$$y = \frac{V}{B}x - \left(\frac{V}{B} - 1\right)x_B$$

graph this

move to where (0.04, 0.04)

$$x = 1$$

graph

$$\frac{S}{F} =$$

$$\frac{2}{3} = \frac{S}{F}$$

$$S = \frac{2}{3}F$$

$$S = \frac{2}{3}(840)$$

$$S = 560 \frac{\text{kmol}}{\text{h}}$$

stripping

$$y = \frac{V}{B}x - \left(\frac{V}{B} - 1\right)x_B$$

$$F + S = D + B$$

$$Fz_F + Sy_S = Dy_D + Bx_B$$

CMO

$$F = B \text{ and } D = S$$

$$Fz_F + Sy_S = Dx_D + Bx_B$$

$$(840(0.44)) + 560(0) = 560y_D + 840(0.04)$$

$$b) \quad 560y_D = 336$$

$$y_D = 0.6$$

b)

$$y = \frac{V}{B}x + \left(1 - \frac{V}{B}\right)y_D$$

$$y = \frac{840}{560}x + \left(1 - \frac{840}{560}\right)y_D$$

$$y = \frac{840}{560}x + \left(1 - \frac{840}{560}\right)(0.6)$$

$$y = 1.5x - 0.3$$

bottoms

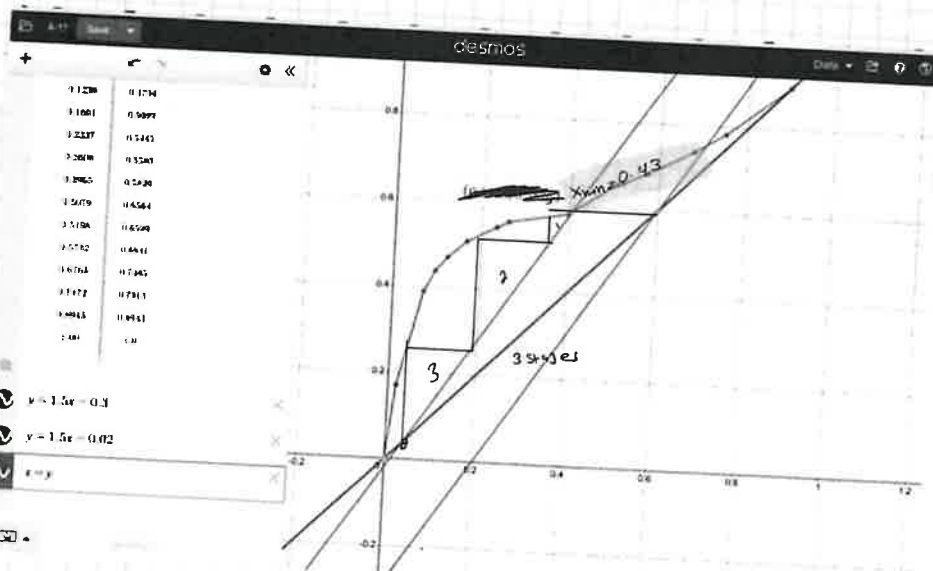
$$y = \frac{B}{S}x + \left(1 - \frac{B}{S}\right)x_B$$

$$y = \frac{840}{560}x + \left(1 - \frac{840}{560}\right)(0.04)$$

$$y = 1.5x - 0.02$$

$$q = 1 \quad K = 0.44 = 2$$

3 eq stages



2

D12. A distillation column is connected to another separation device. The fresh feed to the column is $F = 1000.0 \text{ kmol/h}$ of a saturated vapor that is 20.0 mol% ethanol and 80.0 mol% water. The feed is input at its optimum feed stage. The distillation column has a partial condenser and no reboiler. The column is heated with direct injection of pure steam that is a saturated vapor at 1.0 atm. The ratio bottoms flow rate/steam flow rate $= B/S = 5.0$. The reflux ratio $L_0/D = 1.0$. The distillate from the column is 70.0 mol% ethanol and the bottoms is 1.0 mol% ethanol. The column operates at a pressure of 1.0 atm, the reflux is returned as a saturated liquid, and CMO is assumed to be valid. VLE data are in Table 2-1.

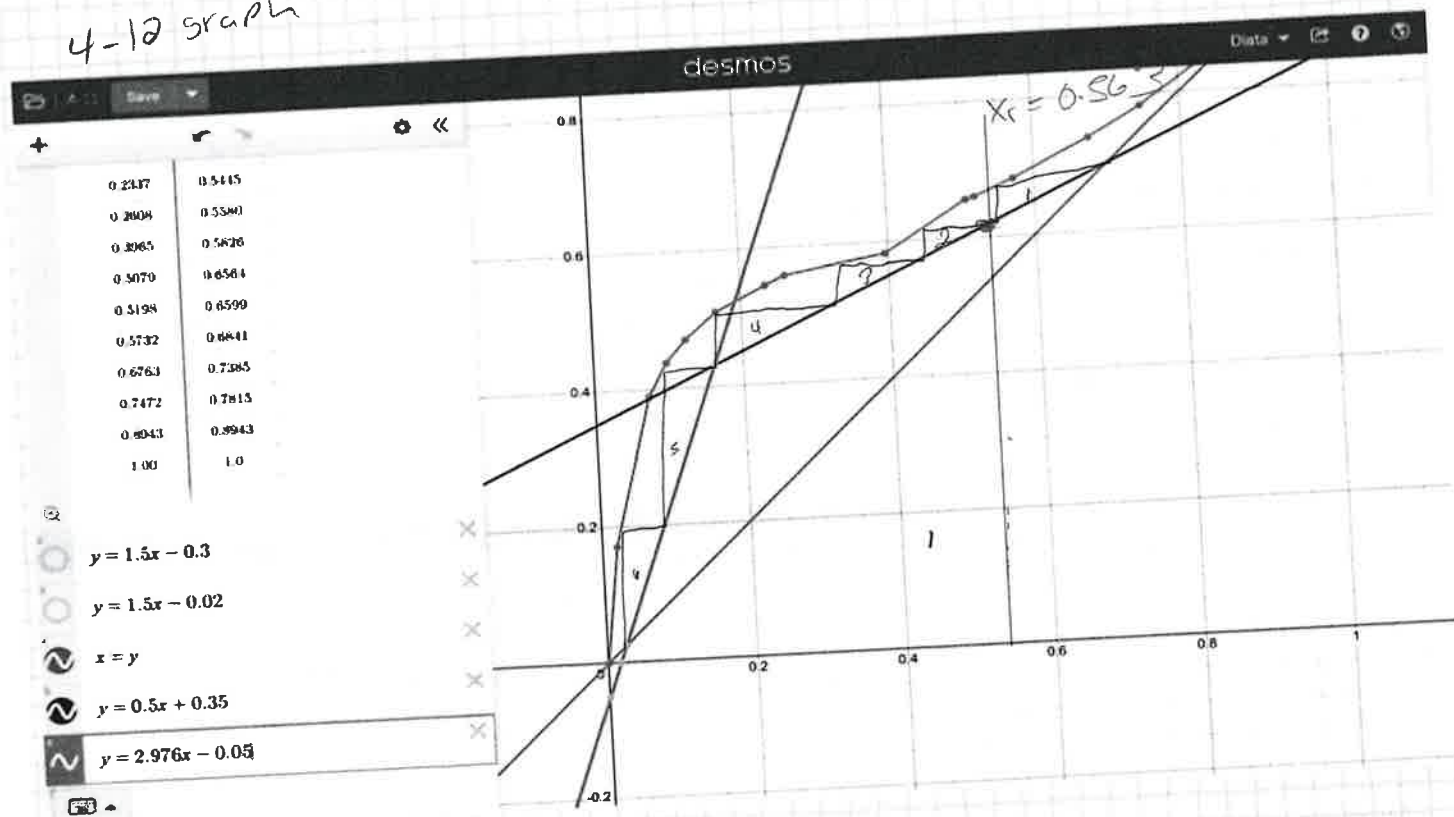
The distillate is sent to a separation device that, because it operates on a different principle than distillation, is not limited by the presence of an azeotrope. The device produces product P and recycle stream R that is a saturated liquid returned to the distillation column at its optimum feed stage. The product P concentration from the device is 98.7 mol% ethanol.

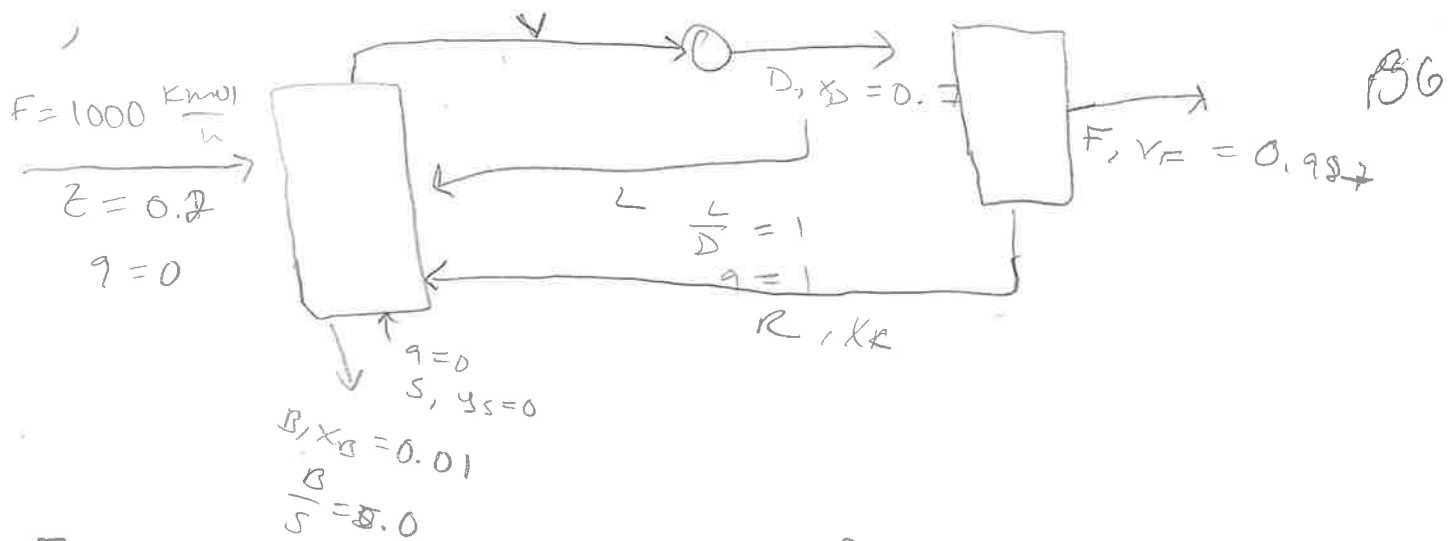
- a. Find steam flow rate S, bottoms flow rate B, distillate flow rate D, product flow rate P, and recycle rate R—all in kmol/h —and find mole fraction of

ethanol in recycle stream x_R .

- b. For the distillation column, find the optimum feed plate for the recycle stream, the optimum feed plate for the fresh feed, and the total number of equilibrium stages required.

4-12 graph





$$F + S = P + B$$

$$Fz + Sy_S = Px_P + Bx_B$$

$$\text{if } \frac{B}{S} = 8 \quad B = 8S$$

$$\text{if } \frac{L}{D} = 1, L = D$$

$$S + F = D + D$$

$$\frac{S + F}{2} = D$$

$$1000(0.2) + S(0) = 0.987P + 0.01B$$

$$200 = 0.987P + 0.01B$$

$$P = 1000 - 0.8B$$

$$a) \quad 200 = 0.01B + 0.987(1000 - 0.8B)$$

$$-787 = -0.7796B$$

$$B = 1009.49 \frac{\text{kmol}}{\text{h}}$$

$$P = 1000 - 0.8(1009.49)$$

$$P = 192.41$$

$$S = 192.41 + 1009.49 - 1000$$

$$S = 201.9 \frac{\text{kmol}}{\text{h}}$$

$$D = \frac{201.9 + 1000}{2} = 600.95$$

$$R = D - P = 600.95 - 192.41$$

$$= 408.54$$

$$F + S = P + B$$

$$1000 + \frac{B}{S} = P + B$$

$$P = 1000 - 0.5B$$

$$Dx_D = Px_D + Rx_R$$

$$x_e = \frac{Dx_D - Px_D}{R}$$

$$= 0.565$$

$$\frac{L}{D} = 1$$

pg 7

Enriching line (TOP)

$$y = \frac{L}{L/D + 1} x + \frac{1}{L/D + 1} x_B$$

$$y = \frac{1}{2} x + (0.5)(0.7)$$

$$y = 0.5x + 0.35$$

stripping

$$y = \frac{D}{S} x + \frac{B}{S} x_B$$

$$y = \frac{600.95}{201.9} x + \frac{1009.49}{201.9} (0.01)$$

$$y = 2.976x - 0.05$$

fresh

$$q=0 \rightarrow y = 2 = 0.2$$

$$q=1 \rightarrow x = x_e = 0.563$$

b)

$x_e = 0.56$ recycled optimum feed plate: 6

Fresh feed plate: 2 from stream

Total # of eq stages: 6